

**NACETEM**

Vol 5 NO 5 | November, 2025

NEWS LETTER

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A Publication of National Centre for Technology Management

From the desk of the DG- Energy Loss in Nigeria — Why the Conversation Must Shift to Systems Thinking



Nigeria's persistent struggle with energy loss has long been treated as a simple technical problem, but global patterns show that countries that successfully solve power-sector inefficiencies begin by understanding the

electricity system as a living, interconnected whole. The conversation is not only about malfunctioning infrastructure or insufficient generation capacity; it is about the complex interplay between generation, transmission,

distribution, regulatory environments, investment incentives, and consumer behaviour. When countries like Denmark, Japan and South Korea drastically reduced their transmission and distribution losses, they did so by adopting a systems-thinking framework that allowed them to view energy not as an isolated commodity but

as a dynamic ecosystem.

In Nigeria, transmission and distribution losses continue to exceed international averages by a significant margin. These losses represent billions of naira in wasted resources every year, but more importantly, they signify lost opportunities for industrial productivity, social stability and investor confidence. Evidence from countries that have undergone grid modernization suggests that technologies alone smart metering, predictive maintenance systems, Internet-of-Things sensors will not deliver sustained improvement unless supported by systems-based policy reforms. Japan, for example, did not achieve single-digit system losses merely by installing smart meters; it redesigned stakeholder roles across

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the utility chain, enforced strict performance standards, ensured transparency in energy accounting, and linked operational behaviour to long-term national planning.

A systems-thinking approach to Nigeria's power sector would highlight the reinforcing feedback loops that keep losses high: insufficient investment leads to ageing equipment; ageing equipment produces higher losses; high losses reduce revenue and discourage new investments; and low investments perpetuate system decay. Breaking this cycle requires a clear understanding of these loops and deliberate interventions at leverage points such as tariff reform, real-time monitoring, enforcement of technical standards, and reliable

regulatory oversight. Countries like Singapore have built world-class energy systems by identifying these leverage points early and sustaining intervention through consistent governance.

Modern energy planning also increasingly integrates artificial intelligence to predict faults, detect leakages, and forecast demand patterns. Nigeria's National Artificial Intelligence Strategy points to the possibility of deploying machine-learning tools across the grid to drastically reduce technical and non-technical losses. Several large power utilities around the world have demonstrated that AI-enabled grid optimisation can cut technical losses by up to 30 percent, representing massive cost savings and significant

improvements in reliability. For Nigeria, where power availability directly shapes national competitiveness, such technological adoption supported by systems thinking offers a viable path toward stability.

Above all, Nigeria needs a policy mindset that recognizes power as the backbone of national development and treats energy loss not simply as an engineering malfunction but as a systemic problem requiring coordinated, multi-sectoral action. With clear political will, strategic investment, and a systems-dynamics-driven reform agenda, Nigeria can reposition its energy sector to support industrial growth, enhance national security, and improve the quality of life for millions of citizens.

Speech by the Permanent Secretary at the G-STIC Conference



The Permanent Secretary, Esuabana Nko-Asanye, stated that the conference provides a vital opportunity to strengthen Nigeria's innovation ecosystem, expand national AI capacity, and embed systems thinking into policymaking. She noted that the G-STIC platform connects Nigeria to a global community working to accelerate the SDGs

through advanced technologies and innovation.

She highlighted that AI is transforming key sectors—agriculture, health, education, energy, and governance—by boosting productivity, improving climate forecasting, enhancing policy efficiency, and creating value in the digital economy. However, she stressed that AI's real

impact depends on adopting a systems-thinking approach that ensures policies are technically sound, inclusive, and environmentally sustainable.

Nko-Asanye reaffirmed the government's commitment to strengthening the national innovation system through evidence-based policies and inter-agency collaboration. She added that the Federal Government is building a robust AI ecosystem via the National Artificial Intelligence Strategy and supporting technology incubation through the NBTI network.

She also noted that institutions such as NACETEM are driving climate action and green innovation, ensuring that Nigeria's STI agenda aligns with global priorities, including the Paris Agreement and the AU's Agenda 2063.



Director General/CEO National Board for Technology Incubation speaking at the Sensitization/stakeholder engagement of the STI Dashboard for Nigeria. The STI dashboard is an ongoing project that explores Nigeria's comprehensive Science, Technology and Innovation (STI) indicator with interactive visualizations. The link is <https://stidashboard.nacetem.gov.ng>



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The Permanent Secretary, Federal Ministry of Innovation, Science and Technology; Cross-Section of Law Makers; The DG/CEO OF NACETEM and THE DG/CEO OF NBTI at the NACETEM 2025 G-STIC Conference



The DG/CEO OF NACETEM



Hajiya Hussaina Tajuddin Abbas with staff of NACETEM AT HE C-STIC CONFERENCE



Mohammed Esat Dizbay (COO OF ECO GREEN INVESTMENT LIMITED UK) facilitating at the NACETEM G-STIC 2025 CONFERENCE



practical Virtual welding @the NACETEM 2025 G-STIC Conference



the wife of the speaker of the house -Hajiya Hussaina Tajuddin Abbas at the NACETEM G-STIC Conference to render her support for NACETEM'S initiatives for standardization and certification of welding in Nigeria



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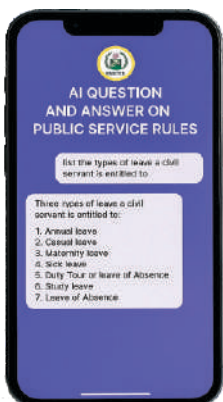
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NACETEM successfully hosted a two-day workshop on the "Digitalization of Content Services for Public Officers" from November 12th-13th, 2025. The event was held at the Nigerian Army Resource Centre in Asokoro, Abuja. The training effectively equipped staff from the National Youth Service Corps (NYSC) and the Financial Reporting Council of Nigeria (FRC) with essential digital skills. This initiative underscores a significant step towards enhancing digital efficiency within Nigeria's public service sector.



Dr Stefano Amenia FACILITATING A TRAINING ON SYSTEMS THINKING AND SYSTEMS DYNAMICS FOR POLICY RESEARCHERS AND POLICY MAKERS ORGANIZED BY NACETEM



NACETEM creates an AI powered Public service rules (PSR) examination preparation app to assist civil servants writing



STI Indicator Dashboard

Thanks to NACETEM's STI Indicator Dashboard, Nigeria now has a unified, transparent, and dynamic tool for tracking its science and innovation ecosystem. This dashboard empowers policymakers, researchers, and industry leaders with real-time data and insights—enabling smarter decisions, stronger collaboration, and a more innovation-led future. Here is the link to the STI Dashboard project: <https://stidashboard.nacetem.gov.ng/>



NACETEM initiates a fruitful collaboration with Public Complaints Commission in areas of electronic citizen's complaints management



Dr Sonnia Agu (CEO OF G1ST INTERNATIONAL FOUNDATIONL CEO OF SAPIO GROUPS) facilitating a training on "turning knowledge to wealth, intellectual capital creation and management in public institutions" organized by NACETEM



Boost your skills with NACETEM's dynamic capacity-building programs, available both virtually and in-person. We offer internationally recognized certifications—including Certiport—alongside flexible self-paced and instructor-led courses. Our trainings typically span one to two weeks, tailored to your needs and the nature of each course. Explore a wide range of topics including ICT, data science, leadership, and management.